PHYSICS WATER COOLER

Ancient Atmospheres

From a Bombarded Ball of Rock to the Air We Breathe

STUDENT STUDY GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

Ancient Atmospheres

This episode follows Earth's atmosphere on its 4.5-billion-year journey. It begins with the birth of the solar system and the violent early Earth, including the collision that made the Moon. It then traces Earth's three atmospheres: the first, of hydrogen and helium, soon lost; the second, built from volcanic gases with essentially no free oxygen; and the third, oxygen-rich atmosphere we breathe today. Along the way, life appears, cyanobacteria invent oxygen-producing photosynthesis, and the oxygen in the third atmosphere rises and falls through six stages, bringing Snowball Earths, the Boring Billion, the Cambrian Explosion, and giant insects.

Your Host

Name	Role
Robert Mahoney	Host; physics and astronomy educator

Learning Objectives

After this episode you should be able to:

- Describe how the solar system formed from a nebula, and how a supernova may have helped trigger its collapse.
- Describe the early Earth of the Hadean Eon, including the heavy bombardment and the Moon-forming collision with Theia.
- Name Earth's three atmospheres, describe the composition of each, and explain why the first one was lost.
- Explain how Earth's oceans formed, and identify three possible sources of Earth's water.
- Describe the earliest life on Earth, and explain why the lack of an ozone layer mattered.
- Explain how cyanobacteria and oxygenic photosynthesis changed Earth, and why the first oxygen never reached the air.
- Describe the Great Oxygenation Event and explain how it led to the first Snowball Earth.
- Name and describe the six stages of oxygen in Earth's third atmosphere, with their approximate dates.
- Explain the balance between photosynthesis and respiration, and describe the composition of today's atmosphere.

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Key Terms

Term	Definition
Supernova	The explosive death of a massive star; its shockwave may have helped trigger the collapse of the cloud that formed our solar system.
Nebula	A vast cloud of gas and dust in space.
Hadean Eon	Earth's earliest eon, from its formation about 4.54 billion years ago, when the planet was hell-like and heavily bombarded.
Planetesimal	One of the small bodies of rock and ice that collided and merged to build the planets.
Heavy bombardment	The early period, lasting hundreds of millions of years, when space rocks pounded the Earth almost continuously.
Theia	The Mars-sized protoplanet whose side-swiping collision with Earth created the Moon.
Tectonic plates	The moving pieces of Earth's crust; today there are roughly seven major and eight minor plates.
First (primordial) atmosphere	Earth's original atmosphere, mostly hydrogen and helium, lost to space early on.
Solar wind	The stream of charged particles flowing out from the Sun; the fierce solar wind of the young Sun helped strip away Earth's first atmosphere.
Second atmosphere	Earth's atmosphere built from volcanic gases: mainly carbon dioxide and nitrogen, with essentially no free oxygen.
Third atmosphere	The oxygen-rich atmosphere we breathe today; it began with the Great Oxygenation Event.
Free oxygen	Oxygen gas (O_2) that is not chemically bound up in rocks, water, or other compounds.
Archean Eon	The eon that followed the Hadean, during which life first appeared and cyanobacteria evolved.
Anaerobic	Living without oxygen, as Earth's earliest microbes did.
Archaea	A group of single-celled microbes; some early archaea consumed carbon dioxide and released methane as a waste product.
Ozone layer	The layer of ozone high in the atmosphere that shields living things from the Sun's ultraviolet radiation.
Cyanobacteria	Bacteria that evolved oxygenic photosynthesis and filled Earth's oceans and air with oxygen.
Oxygenic photosynthesis	Using sunlight to turn carbon dioxide and water into food, releasing oxygen.

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Term	Definition
Banded iron formations	Rusty, layered rock formed when early oxygen reacted with iron dissolved in the oceans; still mined today.
Proterozoic Eon	The eon that followed the Archean, beginning about 2.5 billion years ago, during which oxygen first entered the air.
Great Oxygenation Event	The rise of oxygen in the air about 2.4 billion years ago; it marks the start of the third atmosphere (Stage 1).
Greenhouse gas	A gas, such as methane or carbon dioxide, that traps heat and warms the planet.
Snowball Earth	A time when Earth froze over, from the poles to near or at the equator.
Boring Billion	The long, uneventful stretch of low, steady oxygen that followed the first rise (Stage 2).
Supercontinent	A single landmass made of most or all of the continents; the breakup of one helped drive the Snowball Earths of about 700 million years ago.
Phanerozoic Eon	The current eon, beginning about 540 million years ago, the age of abundant complex life.
Cambrian Explosion	The rapid diversification of complex animal life, about 540 million years ago, when most major animal body plans appear in the fossil record (Stage 4).
Respiration	Taking in oxygen and giving off carbon dioxide; all living things do it, around the clock.
Parts per million (ppm)	A way of expressing very small fractions; today's carbon dioxide is about 425 parts per million, or 0.04 percent.

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Numbers to Know

Value	What it refers to
4.6 billion years ago	A little before this, a nearby supernova helped trigger the collapse of the solar nebula
4.54 billion years ago	Earth formed
4.51 billion years ago	The Moon formed, after the collision with Theia
A few cm per year	How fast the Moon drifts away from Earth
384,000 km (239,000 mi)	The Moon's present average distance from Earth
7 and 8	Today's major and minor tectonic plates
3	Earth's atmospheres: first, second, and third
6	Stages of oxygen within the third atmosphere

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Value	What it refers to
~27 atmospheres	One estimate of the pressure when Earth's surface first cooled below the boiling point of water
Millions of years	How long it rained, off and on, to fill the oceans
More than 70%	Share of Earth's surface the oceans eventually covered
3.8–3.5 billion years ago	When the first single-celled life appeared
4.1–4.2 billion years ago	Debatable hints that life began even earlier
3.4–2.9 billion years ago	When cyanobacteria evolved oxygenic photosynthesis, according to recent genetic studies
2.4 billion years ago	The Great Oxygenation Event; the third atmosphere begins (Stage 1)
Most of a billion years	Length of the Boring Billion (Stage 2)
800 million years ago	Oxygen starts climbing again (Stage 3)
700 million years ago	The second round of Snowball Earths
~10%; 540 million years ago	Atmospheric oxygen at the Cambrian Explosion (Stage 4)
470 million years ago	Plants begin greening the continents
~35%; 290 million years ago	The oxygen peak, the age of giant insects (Stage 5)
~21%	Oxygen today (Stage 6)
~78%	Nitrogen today
0.04% (~425 ppm)	Carbon dioxide today

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The Six Stages of the Third Atmosphere

Stage	What happened
Stage 1 — about 2.4 billion years ago	The Great Oxygenation Event: oxygen enters the air, the first ozone layer forms, and the first Snowball Earth follows.
Stage 2 — the Boring Billion	Oxygen falls back and holds low and steady for most of a billion years.
Stage 3 — from about 800 million years ago	Oxygen climbs again, the ozone layer thickens, and conditions are set for complex life; more Snowball Earths about 700 million years ago.
Stage 4 — about 540 million years ago	Oxygen reaches about 10%; the Cambrian Explosion.
Stage 5 — about 290 million years ago	Oxygen peaks near 35%; the age of giant insects.
Stage 6 — today	Oxygen declines and settles near 21%: the air we breathe right now.

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EPISODE CONTENT

Part-by-Part Summary

Part 1 — The Genesis of the Earth and Moon (The Hadean Eon)

A little over **4.6 billion years ago**, a nearby star ended its life in a **supernova**. Its shockwave helped trigger the collapse of a vast cloud of gas and dust, a **nebula**, and seeded it with heavier elements forged in the dying star. The cloud flattened into a spinning disk with a central bulge that became our **Sun**.

Earth formed about 4.54 billion years ago. Its earliest state was hell-like, almost continuously bombarded by meteoroids, asteroids, comets, and **planetesimals** for hundreds of millions of years. Early on, a Mars-sized protoplanet named **Theia** side-swiped Earth. The debris first formed a ring around Earth, then gathered into the **Moon**, about **4.51 billion years ago**. The Moon began very close to Earth and has been drifting outward a few centimeters a year ever since, to its present average distance of about **384,000 kilometers** (239,000 miles).

Earth's cooling crust would much later fracture into today's moving **tectonic plates**, roughly seven major and eight minor ones, though modern plate tectonics took a long time to get going. Throughout, countless volcanoes spewed gases into the sky.

Earth has had **three atmospheres**. The **first**, or primordial, atmosphere was mostly **hydrogen and helium**, and it escaped: the gases were too light for gravity to hold, or were stripped away by the fierce **solar wind** of the young Sun. A brief atmosphere of vaporized rock followed the Moon-forming impact. What replaced them was the **second atmosphere**, built from **volcanic gases**: mainly **carbon dioxide and nitrogen**, with water vapor, sulfur dioxide, and other gases, and **essentially no free oxygen**. The oxygen-rich **third atmosphere** we breathe today has been anything but stable: its oxygen has risen and fallen through **six stages**.

When Earth's surface finally cooled below the boiling point of water, under a pressure of perhaps **27 atmospheres**, it began to **rain**, off and on, for millions of years. The water was partly native to Earth, possibly partly delivered by Theia, and partly brought later by asteroids and comets. The oceans filled, eventually covering **more than 70%** of the planet.

Part 2 — The Beginning of Life on Earth (The Archean Eon)

The first **single-celled life** appeared sometime between roughly **3.8 and 3.5 billion years ago**, possibly near hot, mineral-rich cracks on the ocean floor. Some researchers see debatable hints of life as early as 4.1 to 4.2 billion years ago.

The air held almost no free oxygen, and therefore **no ozone**, so the Sun's DNA-shredding **ultraviolet** radiation poured onto the surface unchecked. The earliest microbes were **anaerobic**. Some, the **archaea**, consumed the then-abundant carbon dioxide dissolved in water and released **methane** as a waste product.

Sometime in the Archean, likely between about **3.4 and 2.9 billion years ago** according to recent genetic studies, **cyanobacteria** evolved **oxygenic photosynthesis**: using sunlight, they consumed carbon dioxide and water and released oxygen. At first, that oxygen **never reached**

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the air. It reacted with iron dissolved in the oceans, forming rusty layers on the ocean floor, the **banded iron formations** we still mine today. Microbes nearest the surface were killed by ultraviolet light, but layers beneath were shielded by the dead cells above, so life persisted. All this still happened under the second atmosphere.

Part 3 — The Third Atmosphere Begins: Stages 1 Through 3 (The Proterozoic Eon)

By about **2.4 billion years ago**, the oceans' dissolved iron was largely used up, and oxygen began escaping into the air. High up, it formed the **first true ozone layer**. This is the **Great Oxygenation Event**, the arrival of the third atmosphere: **Stage 1**.

Methane had been an important **greenhouse gas**, keeping Earth warm under the young, fainter Sun. Oxygen destroyed that methane, and the planet plunged into a deep freeze, a **Snowball Earth**. Falling carbon dioxide also played a role.

After the first surge, oxygen fell back and held low and steady for most of a billion years, the **Boring Billion**: **Stage 2**. Then, beginning around **800 million years ago**, oxygen started climbing again, a thicker ozone layer formed, and conditions were set for complex life: **Stage 3**. Plants would not green the continents until about **470 million years ago**, with animals following.

Around **700 million years ago**, Earth froze over again in another round of **Snowball Earths**, driven mainly by the breakup of a **supercontinent**, intense rock weathering, and the resulting drawdown of carbon dioxide.

Part 4 — Stages 4, 5, and 6 (The Phanerozoic Eon)

When oxygen climbed to around **10%**, about **540 million years ago**, complex animal life diversified with astonishing speed in the **Cambrian Explosion**, establishing the lineages of most animal life today: **Stage 4**. Simple multicellular life had appeared tens of millions of years earlier, but most major animal body plans first appear in the fossil record in the Cambrian.

Oxygen kept climbing, fitfully, until about **290 million years ago**, when by the best model estimates it peaked near **35%**: the age of **giant insects**, **Stage 5**. Oxygen then declined, for reasons including vast wildfires, settling near today's **21%**: **Stage 6**, the air we breathe now.

Part 5 — The Current Atmosphere, Current Life, and a Forward-Looking Perspective

During the day, plants, algae, and cyanobacteria **photosynthesize**, taking in carbon dioxide and releasing oxygen. But all living things also **respire**, taking in oxygen and giving off carbon dioxide, around the clock. In daylight, plants photosynthesize faster than they respire, so they release oxygen on balance; at night, only respiration remains.

Carbon dioxide, once the dominant gas, is now about **0.04%** of the atmosphere, roughly **425 parts per million**. Nitrogen and oxygen make up almost all the rest, at about **78%** and **21%**. Many atmospheric scientists are concerned that rising carbon dioxide, driven largely by burning fossil fuels, is overheating the Earth, bringing rising seas, disrupted jet streams and ocean currents, and more extreme weather.

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From a hellish, bombarded ball of rock to the only known planet supporting life, Earth's atmosphere has traveled an extraordinary 4.5-billion-year journey through **three great atmospheres**, with **six stages** within the third.

TRY IT YOURSELF

Squeeze Earth's history into one year. If Earth formed at midnight on January 1 and today is midnight on December 31, each day stands for about 12.4 million years. Work out the dates for the Moon's formation (4.51 billion years ago), the first life (about 3.8 billion), the Great Oxygenation Event (2.4 billion), the Cambrian Explosion (540 million), and the oxygen peak (290 million). When do modern humans, about 300,000 years old, arrive?

(Answer: the Moon forms on January 3, the first life around March 1, and the Great Oxygenation Event on June 22. The Cambrian Explosion waits until November 18, and the oxygen peak until December 8. Modern humans show up about 35 minutes before midnight on December 31.)

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STUDY QUESTIONS

Questions for Review

Answer in your own words. Where a question asks you to explain rather than to state, a one-sentence answer is unlikely to be complete.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — The Genesis of the Earth and Moon (The Hadean Eon)

1. What event, a little over 4.6 billion years ago, may have helped start the formation of our solar system? What two things did it do to the nebula?

2. How did the nebula's shape change as it collapsed? What became of the bulge at its center?

3. When did Earth form? Describe its earliest state.

4. How and when did the Moon form?

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5. Where did the Moon begin its life, and how has its distance from Earth changed? What is its average distance today?

6. What did Earth's cooling crust eventually become? How many major and minor pieces are there today?

7. How many atmospheres has Earth had? Name them.

8. What was Earth's first atmosphere made of, and what are the two reasons it was lost?

9. Where did Earth's second atmosphere come from? What was it mostly made of, and what crucial gas was missing?

10. When did it begin to rain on Earth, and for how long? Roughly what pressure was the atmosphere under?

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11. Name three possible sources of Earth's water. How much of Earth did the oceans eventually cover?

Part 2 — The Beginning of Life on Earth (The Archean Eon)

1. When and where did the first single-celled life probably appear? What earlier date has been suggested?

2. Why was there no ozone layer during the Archean Eon? What did that mean for life on the surface?

3. What does “anaerobic” mean? How did the archaea make a living, and what waste gas did they produce?

4. What “trick” did cyanobacteria evolve, and roughly when? What does the process take in and give off?

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5. Why did the first oxygen made by cyanobacteria never reach the air? What evidence of it can we still see today?

6. How did life survive the ultraviolet radiation near the ocean surface?

7. Under which of Earth's atmospheres did all of the events of the Archean Eon unfold?

Part 3 — The Third Atmosphere Begins: Stages 1 Through 3 (The Proterozoic Eon)

1. What happened about 2.4 billion years ago that let oxygen finally enter the air? What formed high in the atmosphere?

2. What is the Great Oxygenation Event? Which atmosphere and which stage does it begin?

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3. Why did the rise of oxygen plunge Earth into a Snowball Earth? What other gas played a role?

4. What is the Boring Billion? Which stage of the third atmosphere is it?

5. What began around 800 million years ago? Which stage is it?

6. When did plants begin greening the continents?

7. When did the second round of Snowball Earths occur, and what three things drove it?

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Part 4 — Stages 4, 5, and 6 (The Phanerozoic Eon)

1. What was the Cambrian Explosion? When did it happen, and about how much oxygen was in the air?

2. Multicellular life existed before the Cambrian. So what is special about the Cambrian?

3. When did oxygen reach its peak, and how high did it get? What kind of animals is this period known for?

4. What happened to oxygen after its peak? Give one reason, and today's oxygen level.

5. List the six stages of oxygen in the third atmosphere, with an approximate date and one key event for each.

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Part 5 — The Current Atmosphere, Current Life, and a Forward-Looking Perspective

1. What do plants, algae, and cyanobacteria do during the day through photosynthesis?

2. What is respiration, and which living things do it? When?

3. Why do plants release oxygen on balance during the day, but not at night?

4. How much of today's atmosphere is carbon dioxide? Express it both as a percentage and in parts per million.

5. What two gases make up almost all the rest of today's atmosphere, and in what proportions?

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6. Why are many atmospheric scientists concerned about rising carbon dioxide? Name three consequences mentioned.

7. Summarize the journey of Earth's atmosphere in one or two sentences, naming its atmospheres and stages.

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YOUR NOTES

Notes

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